Project Name: Food Safety Intelligence Platform

Project Team Members:
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WHY

Problem Statement

Describe the problem you are solving.

Foodborne illness is a major public health issue. The CDC estimates that 48 million Americans get sick from foodborne illness each year, with 128,000 hospitalizations and 3,000 deaths.

Restaurant inspection data is public, but it is hard for everyday people to interpret. Existing inspection records are usually technical, fragmented, and more descriptive. A consumer may be able to see whether a restaurant passed or failed a past inspection, but it is difficult to answer:

Is this restaurant becoming riskier over time, and what factors are driving that risk?

Our project will build a Predictive Food Safety Intelligence Platform that uses Chicago restaurant inspection, 311 complaint, and business license data to predict future food safety risk and explain the key drivers behind that risk. The MVP will focus on Chicago restaurants and present results through an interactive map, restaurant detail page, and explainable risk score.

Why is this a compelling and impactful problem to solve?

Food safety is a major public health issue. Our proposal notes that foodborne illness affects millions of Americans each year, leading to hospitalizations and deaths.

This problem matters because better food safety intelligence could help multiple groups:

- **Consumers** make more informed restaurant choices.
- **Public health agencies** prioritize limited inspection resources.
- **Restaurants** identify operational risks earlier.
- **Platforms and insurers** monitor merchant or business risk.

The impact is both social and practical where the project turns public data into actionable risk information.

Why is this a big opportunity?

Most existing tools show historical inspection results. They do not provide a modern, predictive, explainable risk layer.

The larger opportunity is to create a scalable food safety intelligence platform that could eventually serve:

- cities
- restaurant groups
- food delivery platforms
- POS platforms
- insurers

For the capstone, we will start with a focused Chicago MVP, but the broader opportunity is a reusable risk intelligence layer for food safety.

What assumptions are you making about the problem / opportunity? (these assumptions would directly influence the key elements and features you would build in the minimal viable product (MVP)

- We assume prior violations, failed inspections, repeat issues, inspection frequency, and days since last inspection can help predict future food safety risk.
- We assume 311 complaints, business licenses, neighborhood patterns, and possibly weather can add useful context.
- We assume predictions could affect restaurant reputation, so the product must avoid overclaiming. Use language like “predicted risk” or “risk signal,” not “unsafe restaurant.” Clearly explain model limitations.

Impact and market opportunity

How would you go about quantifying impact, market opportunity for this particular problem you are trying to solve by building a data science product?

Market Size

There is a real market around food safety, compliance, and restaurant operations.
The global food safety testing market was estimated at $26.27B in 2025 and projected to reach $48.01B by 2033.

The restaurant food safety compliance software market is estimated around $1.2B–$1.4B in 2024, depending on the market research source, with forecasts around $3B+ by 2033.

Target Customer and user/customer discovery

Who is the primary customer/user? What is the use case? What are key assumptions you are making about the primary customer/ user and use case?

Identify targeted user/customer segments for the MVP.

Our targeted user is Chicago consumers who want to make more informed decisions about where to eat based on restaurant food safety risk.

Although the product could eventually serve public health agencies, restaurants, platforms, and insurers, the MVP will focus on a public-facing consumer use case because it is easier to prototype and validate within the capstone timeline

Define the primary use case validated by this target user/customer.
The primary use case is when a consumer searches for or browses restaurants in Chicago and uses an explainable risk score to understand whether a restaurant may have elevated food safety risk and what factors are driving that risk.

The target users are the Chicago-area diners who already use digital tools to choose restaurants, such as Google Maps, Yelp, DoorDash, OpenTable, or neighborhood recommendation sites.

What might be other key assumptions that are important to validate with the target user/customer?

- Assumption 1: We assume consumers would consider food safety information if it were easy to understand and presented at the right moment
    - The product should be simple, searchable, and map-based.
- Assumption 2: We assume users find existing inspection records hard to understand.
    - The product should translate inspection history into plain-English explanations and trends.
- Assumption 3: We assume a risk score alone will not be trusted unless users can see the drivers behind it.
    - Each restaurant detail page should show the top contributing factors, such as repeat critical violations, recent failed inspections, complaint density, or time since last inspection.

How would you go about validating these additional key assumptions?

Interview Chicago area consumers/diners. Ask questions like:

- Would food safety risk affect where you eat?
- Do you ever look up restaurant inspection results? Would you look at one if it's easily accessible?
- What would make you trust or distrust a risk score?
- Would you rather see a score, trend, explanation, or map?
- What language feels fair: "risk signal," "predicted risk," "inspection concern," etc.?

Who will you contact to conduct initial user research and feedback?

- Consumers
    - Chicago-area friends and family members
    - People who frequently eat out or order delivery
    - Users who rely on Yelp, Google Maps, DoorDash, or OpenTable
- Subject matter experts
    - Restaurant owner / managers

What is the user journey and UI/UX for this data product?

- User opens the platform
  - They see a Chicago map and search bar. Search bar that could take NLP prompts.
- User searches or browses restaurants
  - They search by restaurant name, address, or neighborhood.
- User selects a restaurant
  - A summary card shows the restaurant's latest inspection result and predicted food safety risk level.
- User reviews key risk drivers
  - The product explains why the restaurant has that risk level, such as repeat violations, recent failed inspections, or nearby complaint trends.
- User makes a more informed decision
  - The user can compare nearby restaurants or view more details before deciding where to eat.

Market Landscape / Competitive Landscape / Existing companies solving the same / similar problem

Who are the major players and main vendors in the space? What are the existing solutions?

Company Name	Stage (startup, enterprise)	Product / Solution overview	Who is the primary customer?	Key differentiation vs your proposal (based on your understanding/ research)
Chicago Food Inspection Forecasting	Government / civic analytics	Predicts likely critical violations to prioritize inspections	Chicago Department of Public Health	Strong precedent, but not a modern public-facing product or startup platform. No SHAP explanations, trend analysis, Yelp NLP.
Yelp health scores	Enterprise	Health inspection scores and health alerts displayed on restaurant listings	Consumers	Display of historical inspection records, no prediction model, no API.
Crunchtime AI	Enterprise	Enterprise restaurant operations software	Restaurant chains	Internal facing for restaurant

		for managing inventory, labor, temperature monitoring, and food safety, and audits.		operations teams. Temperature monitoring. No consumer-facing risk score.
ConnectedFresh	Startup	IoT monitoring platform for restaurants. Sensors track temp, humidity, door status, and energy usage to send real-time alerts and audit ready logs.	Restaurants, grocery stores.	Hardware dependent and internal facing, sensors monitor conditions inside the restaurant. No city data, predictive model, or consumer output.
FoodReady	Startup	AI powered food safety and compliance platform	Food manufacturers, processors, and distributors.	Not targeting restaurant inspection intelligence. Focused on compliance for internal production rather than predicting public health inspection outcomes.
Safe LA Dining	Consumer software	Restaurant health inspection score	Consumers	Pure consumer facing. No ML, a static lookup tool. No API or B2B layer.
Square / Toast / Owner.com	Restaurant SaaS	Help restaurants increase revenue	Restaurants	Mainly serves businesses to increase revenue. No predictive risk model or public health orientation.
SafetyCulture	Enterprise SaaS	Digital audits, checklists, compliance workflows	Restaurants, hospitality, operations teams	Focuses on internal compliance execution, not citywide predictive

				risk from public data.
Inspect2GO	Government software	Inspection, permitting, code enforcement workflows	Local government agencies	Helps manage inspection operations, but not primarily consumer-facing predictive risk scoring.

If you can't identify existing solutions or similar solutions that solve the problem, please explain why there isn't an existing solution.

Relevant readings, market research, white papers, academic research (share title and link)

Initial Chicago Model

- *CDC Foodborne Illness Facts and Stats* - establishes public health importance. (source, Nov 2025)
- *Chicago Food Inspection Forecasting* - precedent for predictive inspection prioritization. (source, code, 2014)
- *Hindsight Analysis of the Chicago Food Inspection Forecasting Model* - independent analysis of Chicago's prior model (source, Oct 2019)
- GovTech coverage of Chicago predictive inspection model - reports that predictive analytics helped identify critical violations earlier. (source, May 2015)

Relevant ML Papers

- *Predicting Food Safety Compliance for Informed Food Outlet Inspections*: A Machine Learning Approach. Predicts non-compliant food outlets in England and Wales. Uses socio-demographic, business type, and urbanness features with **synthetic minority oversampling alongside random forest algorithm**. Identifies 84% of non-compliant outlets in a test set of 92,595. (source, Aug 2021).
- Restaurant Reviews → predictions
    - *Using Text Analysis to Target Government Inspections: Evidence from Restaurant Hygiene Inspections and Online Reviews* (source, July 2013)
    - *Discovering Foodborne Illness in Online Restaurant Reviews* - Uses text classification for the discovery of foodborne illness mentioned in Yelp reviews. Evaluates logistic regression, random forest, SVMs, and decision tree. Identified 10 outbreaks and 8523 reports of foodborne illness associated with NYC restaurants (source, Jan 2018).
    - *Which Restaurants Should You Avoid Today?* - Used support vector machine (SVM) to classify 3.8 million tweets gathered from restaurant visitors in New York

over a 4mo period. Monitor tweets of restaurant patrons for symptomatic language, determining 480 potential cases of food poisoning (source, Nov 2013).

Relevant FDA Resources

- *FDA New Era of Smarter Food Safety* - U.S. food safety regulatory framework. 4 Core Elements aim to create a safer and more digital, traceable food system (source)
    - *New Era of Smarter Food Safety Blueprint* - Discusses technologies, analytics, business models, modernization and values that are the building blocks of the regulatory framework (source, March 2024).
- *Virtual Public Meeting on Data and Technology in the New Era of Smarter Food Safety* - public meeting discussing tech-enabled traceability, real-time recall notifications to consumers, retail food safety). (source, meeting summary, April 2024)

Relevant Model Eval + GenAI Topics

- *Fair Decision-Making for Food Inspections* - studies fairness in inspection scheduling. Geographic fairness analysis, scheduling fairness, inspector bias. Ethical AI angle, fairness-aware ranking systems (source, Aug 2021).
- *Cooking Up Risks: Benchmarking and Reducing Food Safety Risks in Large Language Models*. Evaluates food safety risks in LLMs, introduces domain-specific safety benchmarks, discusses guardrails for food-related AI systems. (source, April 2026)

WHAT

Minimal Viable Product (MVP)

- What is the minimal viable product that you are building that will specifically test the fundamental assumptions you have about the problem and the value of your solution?
    - What are the main features and why?
    - What is the value delivered to your user/customer?
    - What is the key question or questions (max 3) that your target user/customer will be able to answer using your Capstone product?

Our MVP is a Chicago-focused Predictive Food Safety Intelligence Platform that helps consumers understand a restaurant's predicted food safety risk before deciding where to eat. This MVP directly tests our core assumptions that public restaurant inspection data, 311 complaint data, and business license data can be transformed into a simple, explainable, and useful risk signal for everyday diners. The main features will include a searchable restaurant lookup, a predicted food safety risk level, a plain-English restaurant detail page, a risk trend over time, and an explainability layer showing the key factors driving the risk score. These features are important because existing inspection records are public but difficult for consumers to interpret, and our product aims to turn fragmented historical records into a more actionable decision-support tool. The value delivered to users is that they can make faster, clearer, and more informed dining decisions without manually reading technical inspection reports. The key questions users will be able to answer are: **Does this restaurant show elevated predicted**

food safety risk? Is this restaurant's risk improving, worsening, or staying stable over time? What factors are driving this restaurant's predicted risk signal?

- What data science approach would you intend to use for the MVP? (this is NOT UI / UX but technical discussion)

For the MVP, we will use Chicago Food Inspection data as the core modeling dataset and enrich it with relevant public data sources such as Chicago 311 complaints and Chicago business license records. We will engineer features that capture a restaurant's historical and temporal risk patterns, including prior failed inspections, critical violation history, repeat violation counts, total violation counts, inspection frequency, days since last inspection, facility risk category, and nearby complaint density. If feasible, we will also parse the free-text violation column into structured violation categories so the model can learn from richer inspection details rather than treating violations only as a binary signal. The initial modeling task will likely be framed as a supervised classification or ranking problem, where the target is whether a restaurant receives a failed inspection or critical violation in a future prediction window. We would start with interpretable baseline models such as logistic regression, then compare performance against tree-based models such as random forest or gradient boosted trees. To support the explainability layer, we would use feature importance or SHAP values to identify the main drivers behind each restaurant's predicted risk score.

- How would you potentially test the efficacy of the MVP? When would you start testing?

We would test the efficacy of the MVP through both model evaluation and user validation. For model evaluation, we would use historical backtesting by training the model on earlier inspection records and testing whether it can predict later failed inspections or critical violations. We would compare the model against simple baselines, such as using only the most recent inspection result or facility risk category, and evaluate performance using metrics such as precision, recall, ROC-AUC, PR-AUC, and lift in the highest-risk group. For user validation, we would show prototype restaurant pages to Chicago-area consumers or frequent diners and ask whether the risk score, trend, and explanations are understandable, trustworthy, and useful for making dining decisions. We would also seek feedback from at least one subject matter expert, such as a restaurant owner, manager, or food safety/public health expert, to make sure our risk framing is responsible and does not overclaim. We would start testing as soon as we have an initial EDA-backed prototype, even before the final model is complete. Model testing can begin once we have a cleaned baseline dataset and initial engineered features, ideally between weeks 4 and 6 of the project timeline.

What is the key differentiation between your MVP and the existing solutions and/or approaches?

Our MVP differentiates by converting public inspection, complaint, and license data into a predictive and explainable food safety risk layer for each restaurant, rather than simply displaying historical inspection scores or supporting internal compliance workflows. It gives consumers and public-health stakeholders a forward-looking view of risk, plus the main drivers behind that risk, in an interactive product experience.

To make the product more useful for a niche but high-impact use case, we will also support users with compromised immunity - such as older adults, cancer patients, transplant recipients, and others who need safer dining choices. These users will be able to search in simple English language such as "Is this restaurant safe for someone with a weak immune system?" and get a plain-language risk summary.

Key product differentiators

- **Violation text NLP:** Instead of using the violations free-text column only as a binary signal, we will use NLP to extract richer patterns from violation descriptions and uncover more nuanced risk signals.
- **Temporal trend scoring:** The model will answer not just "what is the current risk?" but also "is the restaurant becoming riskier over time?"
- **Explainability layer:** We will use SHAP-based explanations to show the main drivers behind each restaurant's risk score.
- **Fairness checks:** We will test whether the model treats restaurants consistently across geography, neighborhood demographics, and cuisine type, and whether any overprediction is explained by actual violation patterns.
- **Consumer-facing design:** The experience is designed for diners and caregivers, not just inspectors or operations teams.
- **Scalability:** The same framework can later become a risk-intelligence layer for cities, food platforms, restaurant groups, and insurers.

Value Proposition (what value/utility does your project/product provide to your intended users?)

Our MVP helps consumers in any metro area with public restaurant inspection data make safer, faster, and more confident dining decisions by turning difficult-to-read inspection records into a clear, explainable food safety risk signal. Instead of forcing users to interpret raw inspection history, the product shows whether a restaurant's risk appears to be rising and what factors are driving that risk.

The value is especially strong for people who need a simpler way to assess food safety, including users with compromised immunity, older adults, and caregivers. They can search in plain English and quickly understand whether a restaurant may be a better or worse choice for their situation.

This solution is more differentiated because it is predictive rather than purely historical, explainable rather than a black-box score, and consumer-facing rather than built only for inspectors or restaurant operators. By combining inspection history with related public signals, it gives users a more actionable and easier-to-understand view of restaurant food safety risk than static inspection record tools.

Mission Statement

Our mission is to build a predictive food safety intelligence platform that makes public restaurant inspection data easier to understand and act on. We will use inspection, complaint, and business license data to identify elevated food safety risk, explain the key drivers behind that risk, and help consumers and public health stakeholders make more informed decisions.

HOW

Data sets

1. Chicago Food Inspections: Primary public dataset of restaurant inspections in Chicago from 2010-present. Inspections are performed by the Chicago Department of Public Health's Food Protection Program and the data is updated regularly. The data contains results of the inspections, the violations noted, and the risk category of each facility. These attributes will be useful for exploratory data analysis. Some attributes that may make the analysis more difficult are that the dataset contains some duplicate inspection reports, and that in July 2018 there were updates to food inspection procedures, so some of the impacts from these changes may be evident in the data. Additionally the violations are listed as a free-text string which will need to be parsed into structured features.
2. Chicago Business Licenses: Public dataset updated daily containing business license records for businesses and restaurants in the Chicago since 2013, to be joined with the inspection records using the License # / License ID. The dataset contains business names, license numbers, a description of the license, license status, and term start dates and expiration dates. One aspect of the data that may make it difficult for exploratory data analysis is that the dataset contains current active licenses and historical licenses. This may require a deduplication step for renewed licenses.
3. Chicago 311 Service Requests: Public dataset of 311 complaints in Chicago since 2018, containing neighborhood level signals around sanitation, garbage, rodents, and related complaints. An EDA challenge is that the 311 complaints are filed against addresses not businesses, so the latitude and longitude coordinates will need to be used for a spatial proximity join to the Food Inspection data which also contains latitude and longitude coordinates. An aspect of the data that needs to be considered during the exploratory data analysis is that there are some records in the dataset from the previous 311 system in Chicago, indicated by the LEGACY_RECORD column. There are also requests of the

type 311 INFORMATION ONLY CALL which are entered with the address of the City's 311 Center and therefore won't be able to be joined based on spatial proximity to businesses using the latitude and longitude coordinates. The records of both these types may need to be dropped.

4. [NOAA Weather Data]: Public dataset containing weather information, including temperature. The original Chicago model used a 3 day moving average of high temperature as a predictor and found weather to be a significant variable as high temperatures are associated with mechanical failures of food cooling equipment and is a primary source of violations. We might consider extending this to also use the daily minimum temperature to capture the overnight food storage safety risks as well as precipitation, since heavy rain events can correlate with flooding and contamination, rodent activity spikes, and limited staff. The data is joinable via date and geo coordinates.
5. [Yelp Open Dataset]: Public dataset containing reviews for businesses and restaurants. One aspect that may be difficult is to join the data with the inspections data using business name fuzzy matching and address matching. This may be a messy process, but once joined the data can be used for structured features such as average recent rating, review velocity, rating trend, and one-star review rate. The reviews can also be used for NLP features with a text classifier or keyword extraction pipeline. Categories and vocab extracted from inspection violation descriptions can be used to inform the keywords and categories for the review classifier.

Project Management

- What is the role of each member (who will do what specifically)?
- Who is the project manager and chief facilitator (and tie breaker) Jun / Arun
- Who is the resident SME? All of us will become the SME
- Who is the product manager? Jun and everyone
- Who is the lead on infrastructure and data engineering? Deepak Infra / Arun DE
- Who is the lead on EDA? - Aurelia / Arun
- Who is the lead on model evaluation? - Bella / Jun
- Who is the lead machine learning engineer? - Deepak / Bella
- Who is the lead MVP application developer? Aurelia / Jun
- Who are the backups to key roles? Listed next to most positions after slash

	Role and responsibilities (immediate)	Role and responsibilities (long term) / Alternate or additional role / pair
Jun Xu	Project manager & Product manager	MVP App and all other areas
Deepak Srivastava	Infrastructure & MLE	Infra and all other areas

Bella Davies	MLE & Eval	EDA and all other areas
Aurelia Yang	EDA & MVP App	Model and all other areas
Arun Agarwal	DE & MLE	DE and all other areas

**Some teams have used pairing principles to assign two members of the team to main tasks.

- What are the strengths and weaknesses of each team member?

	Strengths	Weaknesses
Jun Xu	Analytics, Data Pipeline, Data Wrangling, EDA	Have less experience in deploying the model (offline to online)
Deepak Srivastava	AWS certified SA and AI/ML engineer.	No experience with UI/UX frameworks.
Bella Davies	AI/ML, Evaluation	Less experience with deployment
Aurelia Yang	Data Analytics, Machine Learning, EDA	Less geospatial mapping experience/less knowledge of food safety as a domain
Arun Agarwal	Data Engineering, Machine Learning Engineering, Prompt Engineering, EDA, AWS Certified Cloud Practitioner	UI/UX frontend design; limited prior food safety domain knowledge

Technical Approach and Planning

What methodologies would you use for initial data exploratory analysis to ensure your datasets are sufficient and meaningful?

EDA will proceed in three phases: dataset-level profiling, cross-dataset linkage, and modeling readiness checks.

Phase 1: dataset-level profiling. For each of the five public sources (Chicago Food Inspections, Chicago Business Licenses, 311 Service Requests, NOAA Weather, Yelp Open Dataset), we will check shape, schema, time coverage, missingness, duplicate rates, and outliers. We will pay specific attention to the documented quirks already called out in the Datasets section, including the July 2018 inspection procedure change, the legacy 311 records, and the 311 information-only entries logged at the City's 311 Center address. For the inspection data, we will produce baseline distributions of inspection results (Pass, Pass with Conditions, Fail) over time, by neighborhood, and by facility risk category. We will also profile the free-text violation column to estimate how many distinct violation patterns exist and how much structure we can reasonably parse out of it.

Phase 2: cross-dataset linkage. The MVP depends on joining inspections to business licenses (by License Number), inspections to 311 complaints (by spatial proximity using latitude and longitude), inspections to NOAA weather (by date and geography), and inspections to Yelp reviews (by fuzzy name and address matching). For each join, we will measure match rate, false match rate on a sampled audit, and the impact of unmatched records on downstream label distributions. The Yelp fuzzy match in particular needs careful evaluation, since incorrect matches will inject systematic noise into the model.

Phase 3: modeling readiness checks. Before any model training, we will confirm that the features we expect to be predictive are actually correlated with the target. This includes prior failed inspections, repeat violation counts, days since last inspection, complaint density, and weather signals. We will also produce a time-aware train/test split (training on earlier years, evaluating on later years) so that model evaluation reflects real-world forecasting conditions rather than a random shuffle that would leak future information.

What data science algorithms are you intending to develop and build for the project? What challenges do you potentially foresee?

The system has four modeling components (as of now):

1. Risk classification or ranking model. Supervised model predicting whether a restaurant will receive a failed inspection or critical violation in a forward-looking prediction window. We will start with logistic regression as an interpretable baseline, then compare against gradient boosted trees (XGBoost or LightGBM) and a random forest. Evaluation uses precision, recall, ROC-AUC, PR-AUC, and lift in the top-decile risk group. Class imbalance will likely require SMOTE or class-weighted training.
2. Violation text classification. NLP layer to parse the free-text violation column into structured violation categories. We will start with TF-IDF and logistic regression as a baseline, then potentially evaluate a fine-tuned transformer and an LLM-based zero-shot classifier (Claude or Llama via AWS Bedrock) for richer category extraction. The output feeds back into the risk model as structured features.
3. Yelp review signal extraction. Conditional on the Yelp join reaching a reasonable match rate, we will extract review-derived features such as rating trend, review velocity, one-star rate, and food safety keyword presence (sick, sanitation, dirty, foodborne, and related terms) using either a fine-tuned classifier or a prompted LLM.

4. Explainability layer. SHAP analysis to identify the top drivers behind each restaurant's predicted risk. The output of this layer is what gets surfaced in the consumer-facing detail page, so it has to be both technically accurate and presentable in plain English.

In addition to the four model components, we will need to run a fairness audit comparing model performance across neighborhood, demographic, and cuisine groupings to check whether the model over-predicts risk in any group beyond what actual violation rates would justify.

Anticipated challenges:

- Data linkage quality. The Yelp fuzzy match and the 311 spatial join are both noisy operations. If match rates are low or false-match rates are high, the model will inherit systematic bias.
- Free-text violation parsing. The violation column is inconsistent across inspectors and over time. Producing clean, structured categories will require iteration.
- Temporal leakage. Forecast accuracy is sensitive to how we set up the train/test split. Restaurants reinspected frequently can leak label information across the cutoff if features are constructed carelessly.
- Class imbalance. Failed inspections and critical violations are a minority class, and standard accuracy will be misleading.
- Responsible-AI framing. Our outputs can affect restaurant reputation. We will need careful language ("predicted risk signal," not "unsafe restaurant"), confidence indicators on the score, and a completed fairness audit before any consumer-facing deployment.
- Deployment on AWS. The MVP requires the model, the explainability layer, and the frontend all to be hosted on AWS within the project's AWS credit budget. Selecting the right Bedrock model for the NLP components and minimizing inference cost is its own optimization problem.

What help do you need?

- A subject-matter expert in food safety or restaurant operations, ideally someone who has worked in or with a public health inspection program, to validate that our risk framing is responsible and that the features we are engineering are reasonable. (also a requirement for the project)
- Feedback on the UI and the consumer-facing risk score language, where the framing has to balance informativeness against not over-claiming.
- Once we get to this point, guidance on AWS Bedrock pricing and model selection for the NLP components, especially if we adopt an LLM for violation text classification or Yelp review signal extraction.
- Connections to Chicago-area diners and at least one restaurant owner or manager for the user research interviews (if we decide to do this)

Note: there is a weekly team check in at the beginning of each class starting week 4 and during non-presentation weeks. The team check-in is usually 3-5 min. Please cover: major milestone(s) achieved in the past week, major milestones to be achieved this week. One key learning to share with the class. And help needed.